

Empowering Head-up AR: Leveraging Holographic Display Engine, Geometry Optical Combiner, and Learned Calibration

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Figure 1: Conceptual illustration of a holographic HUD that applies to windshield AR navigation scenarios and supports multi-plane display experience, leveraging an off-the-shelf free-form geometry optical combiner, which can be compensated by a learning-empowered, calibrated hologram generation algorithm to deliver virtual 3D scenes without distortion.

Abstract

Holography, enabling next-generation Augmented Reality Head-Up Displays (AR-HUDs) through precise wavefront control of light, offers advantages including natural depth perception, enhanced visual comfort, aberration correction, and reduced power consumption. However, widespread adoption of holographic displays, particularly in AR scenarios incorporating optical combiners, has been hindered by persistent challenges in image quality. This work demonstrates how the state-of-the-art learned, camera-calibrated Computer-generated Holography (CGH) algorithms can mitigate this limitation, delivering compelling holographic imagery on a prototype AR-HUD. Crucially, the system employs an off-the-shelf freeform optical combiner, providing adaptability to diverse windshield shapes to facilitate future practical automotive integration.

Keywords

Holography, Head-up Display, Augmented Reality, Calibration

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1 Introduction

Head-up displays with augmented reality (AR-HUDs) exhibit great potential for various applications including entertainment, education/training, medical visualization, and automotive/robotic assistance. Although existing research [Deshpande et al. 2013; Olexa et al. 2024] has made encouraging progress in developing see-through HUDs for 2D content, the lack of 3D cues hinders the immersive depth perception and spatial intuition critical for applications demanding realistic interaction or precise spatial judgment. In contrast to multi-view, light field, and volumetric displays, holographic displays provide depth control of 3D content along with inherent benefits in aberration correction, compact form factor, and high dynamic range [Xia et al. 2025].

Embedding holographic light engines into AR-HUDs typically requires an optical combiner to blend virtual image with real-world scene (Figure 1). Recent work utilizing diffractive waveguide combiners has achieved state-of-the-art imagery quality and compelling 3D depth cues [Gopakumar et al. 2024]. Notably, these solutions can encounter challenges including high cost, complex manufacturing,

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and reduced energy efficiency. Holographic displays with geometric combiners can offer both compact form factor and low cost [Wakunami et al. 2016]. However, distortion calibration still requires tedious algorithms or hardware [Hiroi et al. 2022; Jeon et al. 2023]. In light of these, we seek to leverage a learned, camera-calibrated wave propagation model to compensate for optical aberrations induced by the curved windshield-type optical combiner to deliver 3D content in reasonably high fidelity. Our main contributions include:

- We present an AR-HUDs with the geometry optical combiner for lightweight and cost-effective 3D holographic displays.
- We explore a set of CGH algorithms to calibrate the wave propagation and geometric distortion for AR-HUDs.
- We validate the proposed method in experiments with a windshield-type optical combiner, allowing seamless integration of holography across various optical see-through AR platforms.

2 Method and Results

2.1 HUD Configuration

Figure 2 illustrates our holographic AR-HUD prototype and experimental results. The prototype leverages a phase-only SLM (HDSLM80R-Plus) operated at 60 Hz with the $8\mu\text{m}$ pixel pitch. The SLM has a resolution of $1,920 \times 1,080$ with a bit depth of 10 bits. We utilize a single-color (Green) laser operated at a wavelength of 524.9 nm. All results were captured with a FLIR Grasshopper3 color USB3 vision sensor through a Canon EF 50 mm lens, which can refocus at different depths. We utilize a Nikon 50 mm DSLR lens as the eyepiece, which yields an eye box of 3.28 mm in both directions, following established settings [Peng et al. 2020]. Additional optical components are utilized, including a polarized beam splitter, collimating lens, polarizer, iris, and neutral density filter.

To verify depth cues of holography, we optimize a 3-depth phase-only hologram, which can focus on the near, middle and the far plane. The 2 real objects (character and pumpkin) are placed at near and far depths, same as the virtual planes. Captured images at 3 these depth planes demonstrate nearly aligned defocus effects that are consistent with the real scene.

2.2 Calibrated Hologram Generation

Compared to VR holographic displays, the geometry optical combiner adopted in AR-HUDs introduces extra aberrations and distortions. To calibrate these optical aberrations and other SLM imperfections, we apply the CITL framework [Peng et al. 2020] with the established stochastic gradient descent (SGD) algorithm to optimize phase-only holograms, minimizing losses between captured and target RGB-D images. This can effectively reduce aberrations introduced by curved combiners, especially distortion around edges.

To further model the diffraction induced by the combiner in addition to the geometric distortion, we model the windshield's phase delays using Zernike polynomials with learnable parameters. Additionally, we use two CNNs to construct this learned forward model: the first to model the imperfections of the light source and SLM, while another to model the lens aberrations of the eyepiece and the camera lens.

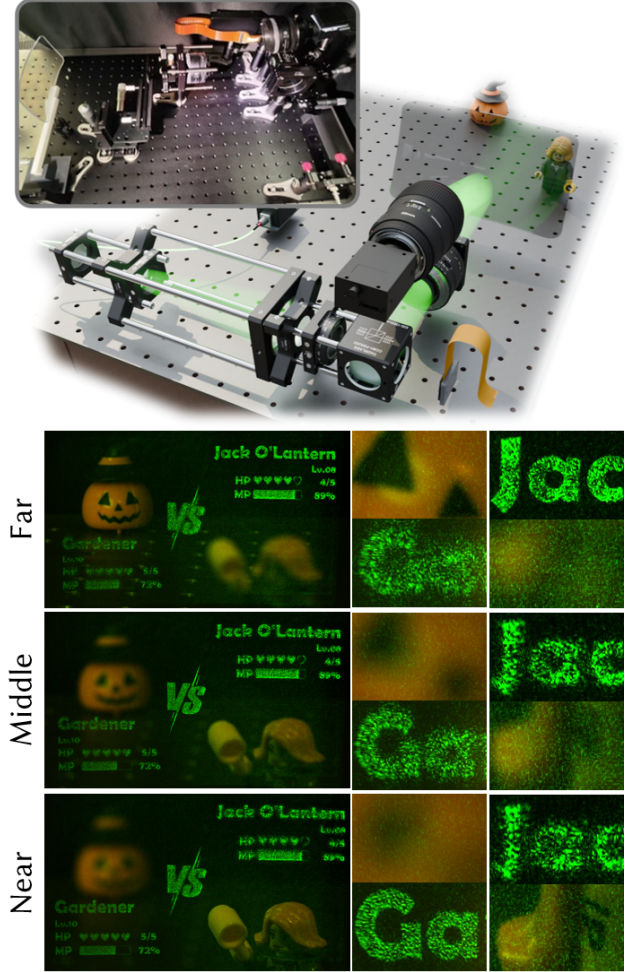


Figure 2: Top: photograph and illustration of the prototype. Bottom: captured holographic display results on three depth planes. Zoomed-in patches reveal in-focus/defocus effects for real and virtual scenes across depths.

We train the forward model on a dataset consisting of holograms and their captured images at the proposed prototype. The holograms contain natural images and common optical phase functions that are generated by various CGH methods to prevent overfitting. The SGD method is then adopted with the trained forward model to reduce geometric distortion while improving image contrast. To accelerate hologram generation in real-time, we can further adopt the neural network-based inverse model [Zhou et al. 2025].

2.3 Summary

In this work, we demonstrate a set of learned, camera-calibrated CGH algorithms for holographic AR-HUDs, built upon off-the-shelf geometric optical combiners. Beyond displaying the captured scenes on a monitor, we will set up an additional beam splitter for users to directly view the output of this AR-HUDs prototype. In such a way, users can perceive 3D content through the windshield combiner projected at designated depths, experiencing holographic displays with nearly true depth perception using their own eyes.

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